THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 9: Differential Equations (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____
Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	The order and degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^5 = \frac{d^2y}{dx^2}$ are: (A) order 1, degree 1 (B) order 1, degree 2 (C) order 2, degree 1 (D) order 2, degree 2	1
2.	The degree of the differential equation $\frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$ is: (A) 1 (B) 2 (C) 0 (D) not defined	1
3.	The integrating factor of the differential equation $x\frac{dy}{dx} - y = x^2$ is: (A) x (B) $-x$ (C) $\frac{1}{x}$ (D) $\log x$	1
4.	The number of arbitrary constants in the general solution of a differential equation of order 4 is: (A) 0 (B) 2 (C) 3 (D) 4	1
5.	The general solution of $\frac{dy}{dx} = e^{x+y}$ is: (A) $e^x + e^{-y} = C$ (B) $e^x + e^y = C$ (C) $e^{-x} + e^y = C$ (D) $e^{-x} + e^{-y} = C$	1

Q No.	Question	Marks
6.	**Assertion (A):** $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 4y = 0$. **Reason (R):** A function is a solution of a differential equation if it satisfies the equation identically. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Find the general solution of $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.	2
	LEVEL 2 **Q 8 – Q 14**	**18**
8.	A bacteria sample grows exponentially. The differential equation representing the growth is $\frac{dP}{dt} = kP$, where P is the population at time t. Obtain the general solution and express P as an exponential function of t.	2
9.	Find the particular solution of $\frac{dy}{dx} = y\tan x$, given that $y = 1$ when $x = 0$.	3
10.	Solve: $x\frac{dy}{dx} + 2y = x^2$ $(x \neq 0)$.	3
11.	Solve the homogeneous differential equation $x\frac{dy}{dx} = x + y$.	3
12.	Verify that $y = Ax + \frac{B}{x}$ is a solution of $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$.	2
13.	Find the general solution of $\frac{dy}{dx} = \sin^{-1}x$.	2
14.	Find the particular solution of the differential equation $\frac{dy}{dx} - 2xy = 3x^2e^{x^2}$, given $y(0) = 5$.	3
	LEVEL 3 **Q 15 – Q 20**	**25**
15.	Solve: $x\,dy - y\,dx = \sqrt{x^2 + y^2}\,dx$.	5
16.	Find the particular solution of $(1 + x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$, given $y = 0$ when $x = 1$.	3
17.	Find the particular solution of $x\sin\left(\frac{y}{x}\right)\frac{dy}{dx} + x - y\sin\left(\frac{y}{x}\right) = 0$, given $y = \frac{\pi}{2}$ when $x = 1$.	5
18.	Solve the differential equation: $x^2\,dy + y(x + y)\,dx = 0$.	3
19.	Solve: $(\tan^{-1}y - x)\,dy = (1 + y^2)\,dx$.	5
20.	**Case Study:** In a laboratory culture, the number of bacteria P increases at a rate proportional to the number present. The count of 1,00,000 bacteria increases by 10% in 2 hours. **(i)** Write the differential equation modelling this situation and state its order and degree. **[1]** **(ii)** Solve it to express P in terms of t and find the constant k. **[1]** **(iii)** In how many hours will the count reach 2,00,000? **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 9: Differential Equations

Q No.	Answer / Key Steps
	LEVEL 1
1.	(C) order 2, degree 1
2.	(D) not a polynomial in derivatives
3.	(C) $\frac{dy}{dx} - \frac{y}{x} = x$, IF $= e^{-\log x} = \frac{1}{x}$
4.	(D) 4
5.	(A) $-e^{-y} = e^{x} + C \Rightarrow e^{x} + e^{-y} = C$
6.	(A) Both true; R explains A ($y'' = 4e^{2x}$).
7.	$\tan^{-1} y = \tan^{-1} x + C$
	LEVEL 2
8.	$\frac{dP}{P} = k\,dt \Rightarrow \log P = kt + C \Rightarrow P = P_0 e^{kt}$ (where $P_0 = e^{C}$)
9.	$\log y = \log \sec x + C$; $y = \sec x$
10.	IF $= x^2$; $yx^2 = \frac{x^4}{4} + C \Rightarrow y = \frac{x^2}{4} + \frac{C}{x^2}$
11.	$y = vx$: $x\frac{dv}{dx} = 1 \Rightarrow y = x\log\lvert x\rvert + Cx$
12.	$y' = A - \frac{B}{x^2}$, $y'' = \frac{2B}{x^3}$; LHS $= \frac{2B}{x} + Ax - \frac{B}{x} - Ax - \frac{B}{x} = 0$
13.	$y = x\sin^{-1} x + \sqrt{1 - x^2} + C$
14.	IF $= e^{-x^2}$: $ye^{-x^2} = x^3 + C$; $C = 5$; $y = (x^3 + 5)e^{x^2}$
	LEVEL 3
15.	$y = vx$: $\frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x} \Rightarrow y + \sqrt{x^2 + y^2} = Cx^2$
16.	IF $= 1 + x^2$: $y(1 + x^2) = \tan^{-1} x - \frac{\pi}{4}$
17.	$y = vx$: $\sin v\,dv = -\frac{dx}{x} \Rightarrow \cos\frac{y}{x} = \log\lvert x\rvert + C$; $C = 0$: $\cos\frac{y}{x} = \log\lvert x\rvert$
18.	$y = vx$: $\frac{dv}{v(v+2)} = -\frac{dx}{x} \Rightarrow \frac{v}{v+2} = \frac{C}{x^2}$; solution $x^2 y = C(y + 2x)$
19.	$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1} y}{1+y^2}$, IF $= e^{\tan^{-1} y}$: $x = \tan^{-1} y - 1 + Ce^{-\tan^{-1} y}$
20.	(i) $\frac{dP}{dt} = kP$; order 1, degree 1. (ii) $P = 100000\,e^{kt}$, $k = \frac{1}{2}\log 1.1$. (iii) $e^{kt} = 2 \Rightarrow t = \frac{2\log 2}{\log 1.1}$ hours (≈ 14.5 h)